



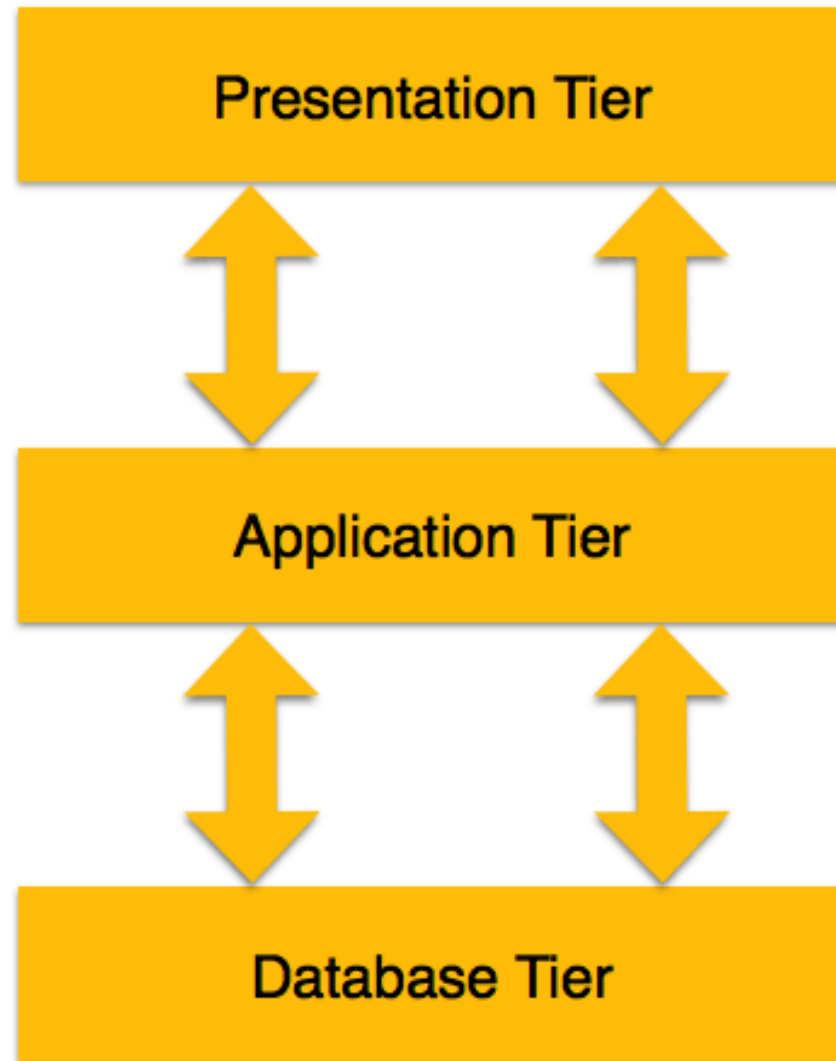
DATABASE-SYSTEMS

Lecture - 2

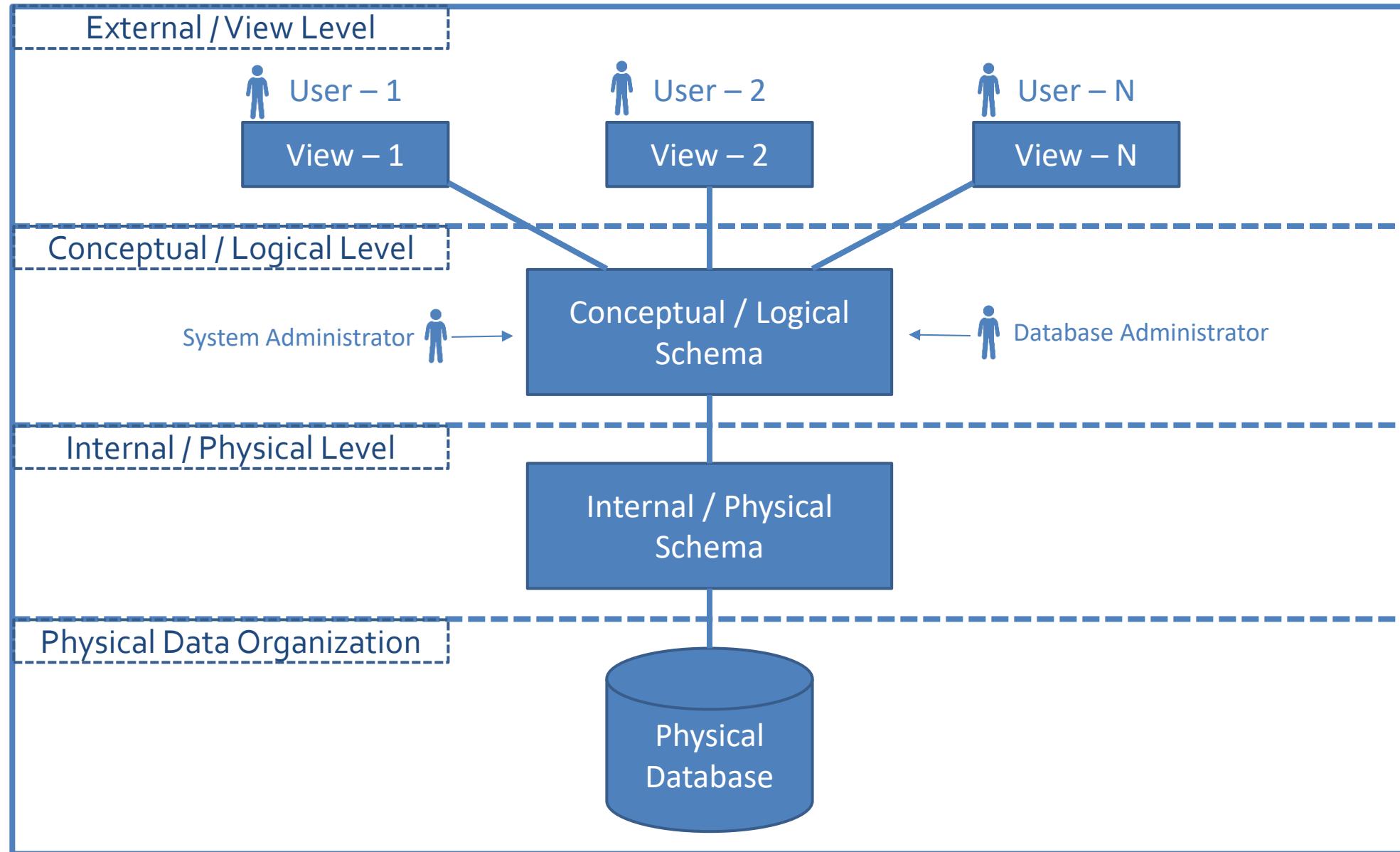
By

M. Qasim Riaz

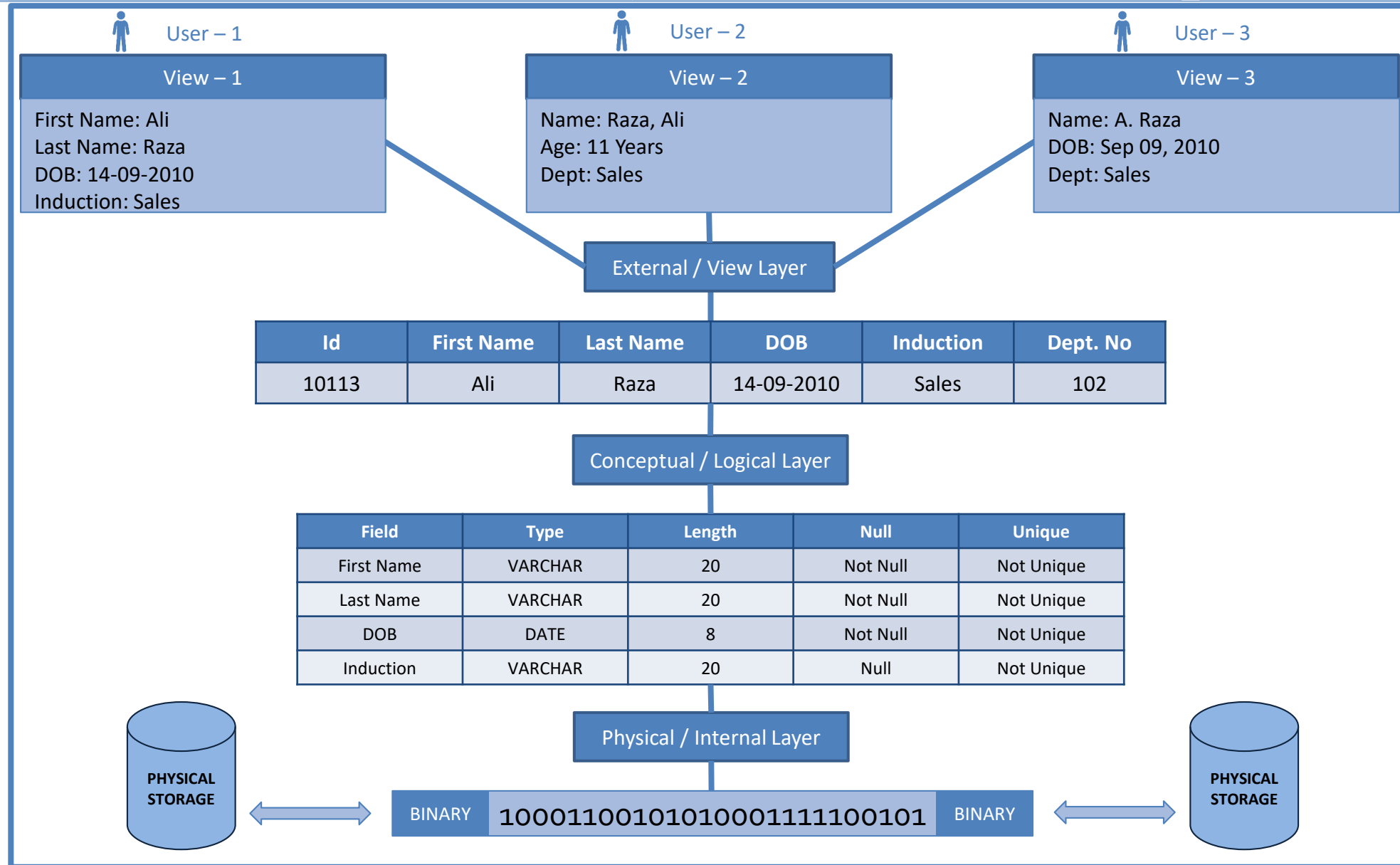
Three Level Architecture (Any Database Application)



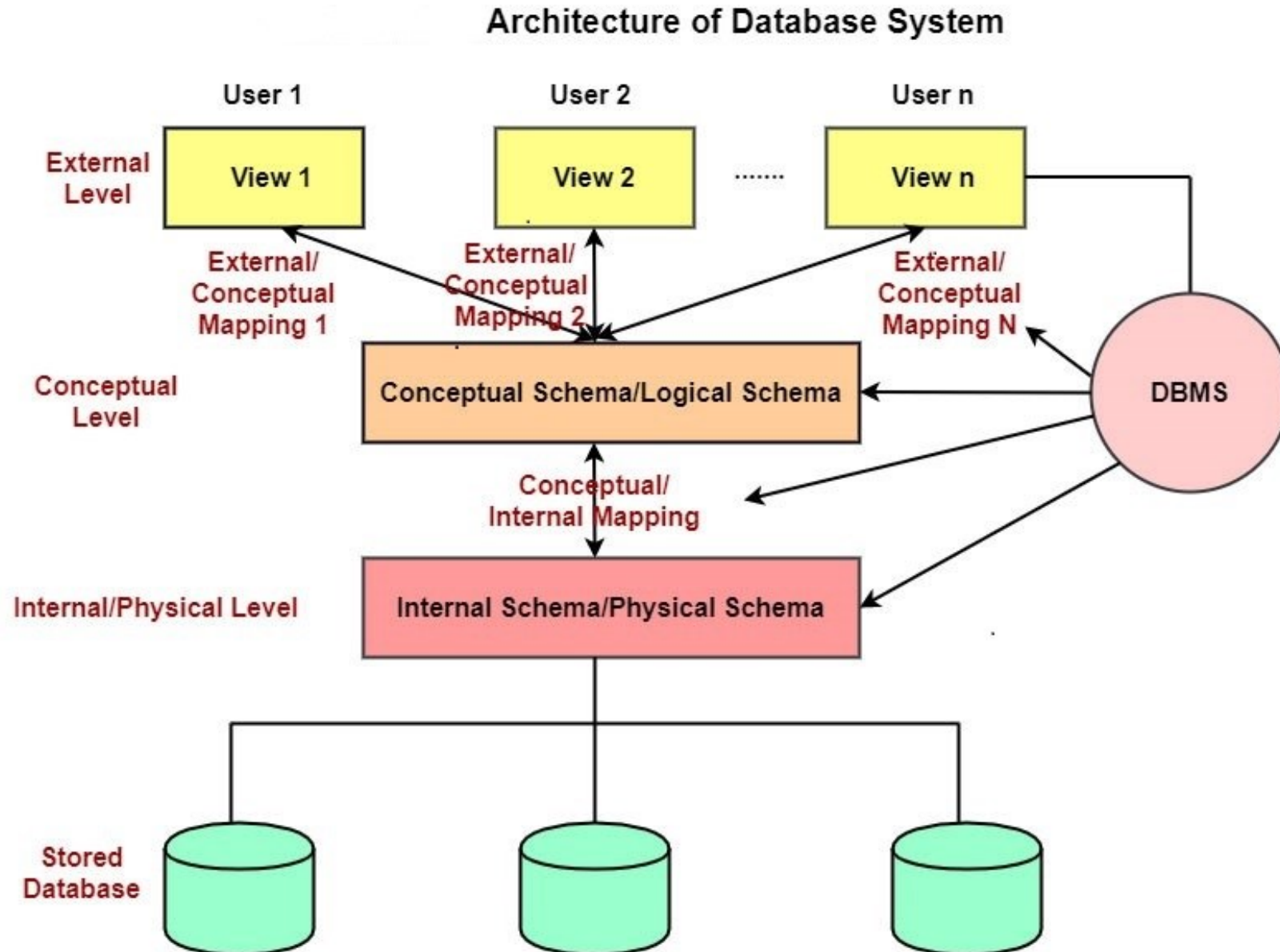
Three Level Architecture (Overall View)



Three Level Architecture(External & Conceptual Level)



Three Level Architecture (Role of DBMS in 3 Levels)



Three Level Architecture (In Simple Words)

External Schema

#	Name	Score
1	sameed	2/3
2	shahzeb	2/3
3	ali	3/3
4	Maria	3/3
5	Imran	1/3

<https://T4Tutorials.com>

Server: 127.0.0.1 » Database: quizdb » Table: questions

#	Name	Type	Collation	Attributes	Null	Default	Extra
1	<u>id</u>	int(11)			No	None	AUTO_INCREMENT
2	question	varchar(300)	latin1_swedish_ci		Yes	NULL	
3	op1	varchar(200)	latin1_swedish_ci		Yes	NULL	
4	op2	varchar(200)	latin1_swedish_ci		Yes	NULL	
5	op3	varchar(200)	latin1_swedish_ci		Yes	NULL	
6	answer	varchar(200)	latin1_swedish_ci		Yes	NULL	

Conceptual Schema

Server: 127.0.0.1 » Database: quizdb » Table: user

#	Name	Type	Collation	Attributes	Null	Default	Extra
1	<u>id</u>	int(11)			No	None	AUTO_INCREMENT
2	name	varchar(60)	latin1_swedish_ci		Yes	NULL	
3	score	varchar(11)	latin1_swedish_ci		Yes	NULL	

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Physical Schema



Data Independence

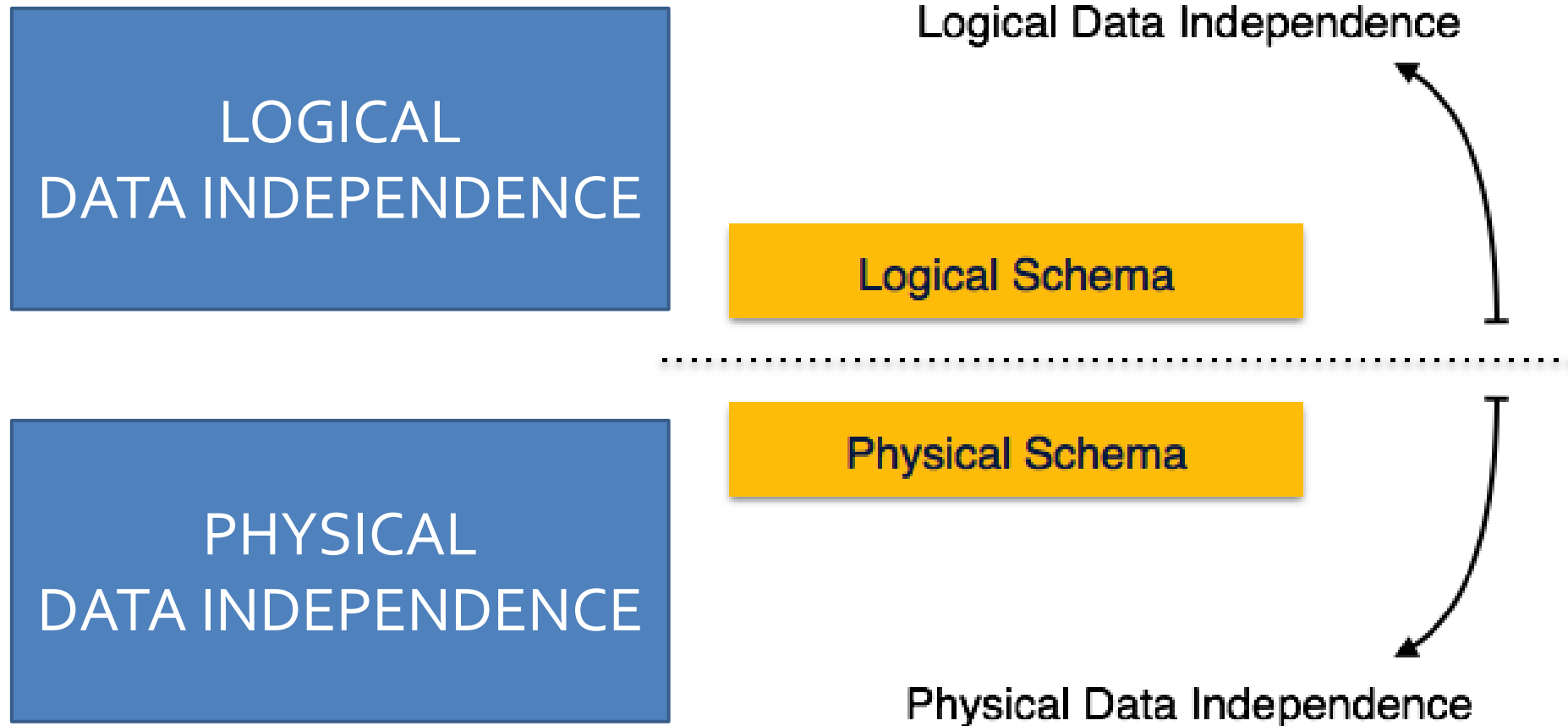
- The **separation of data and application program** is called as data independence.



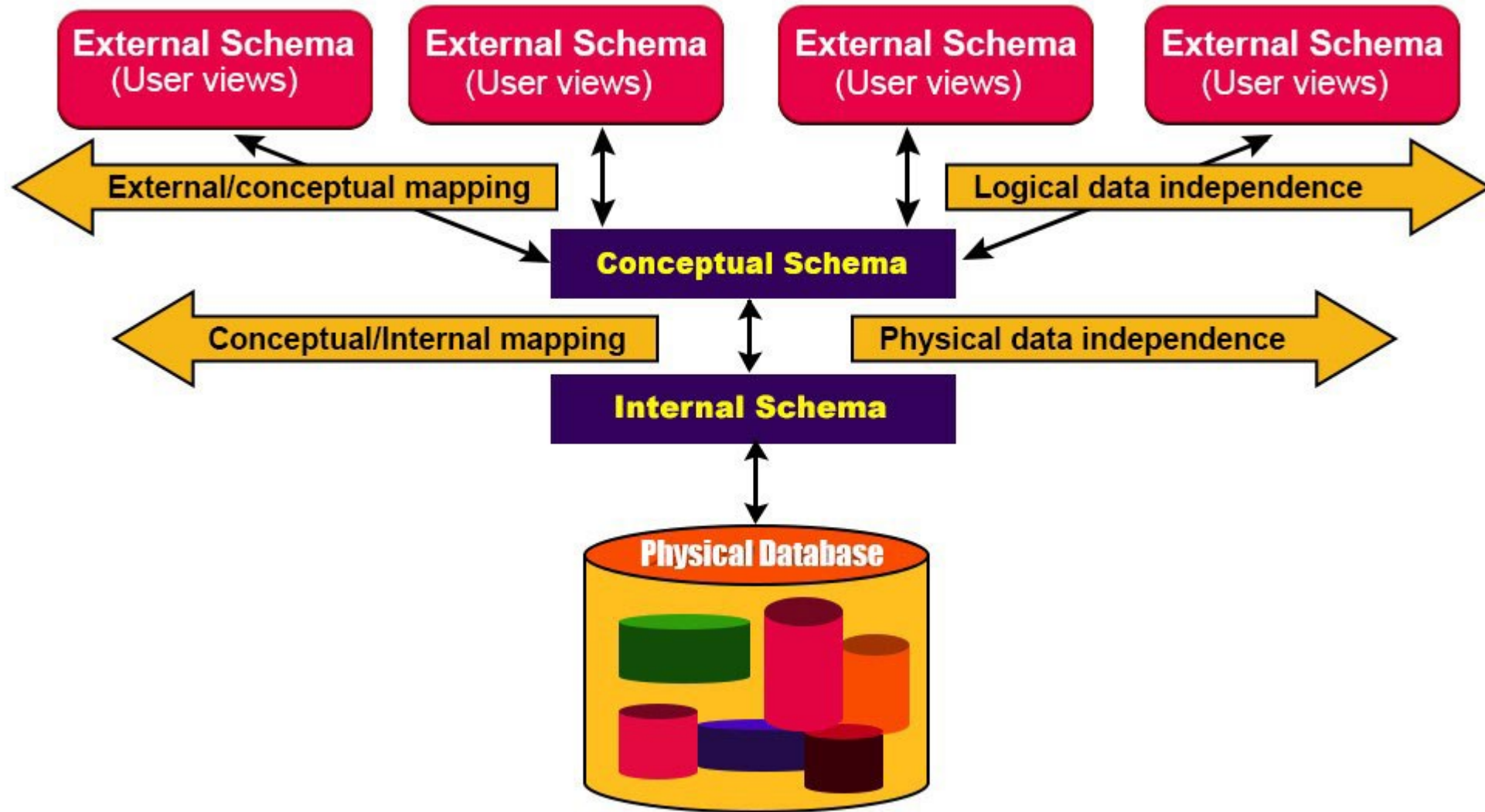
- It is an important feature of DBMS. It is the most **important advantage of three level architecture.**
- Another major advantage is **that any change in lower level architecture does not affect the structure or functionality** of upper levels.
- Data independence **enables the user to change the structure of a database without changing the application programs** or the way users access the data.

Data Independence

- Data independence can be classified into two type based on the level at which the independence is obtained.



Data Independence



Data Independence

PHYSICAL DATA INDEPENDENCE

- Type of independence that **enables the user to change the internal level/physical level without changing the conceptual level/logical level.**
- In DBMS, a **physical structure of database may change without changing application programs or altering the user's view of data** i.e. if the physical structure changes, DBMS is aware of these changes, but still provides the same logical view. The changes that may be performed at physical level without changing the logical level view are as follows:
 - Changing file organization or storage structures
 - Using different storage devices
 - Modifying indexes
 - Changing the access method
 - Changing data structure



Data Independence

- **Logical Data Independence**

- Logical data independence is a **type of independence that enables the user to change the conceptual level/logical level without changing the external level/view level**. The changes that may be performed at logical level without changing the external level are as follows:
 - Adding or removal of entities/tables or relationships
 - Adding a field to the database
 - Removing a field from database



Data Dictionary

- A data dictionary contains metadata i.e. data about the database
- Data dictionaries store all the various schema, file specifications and their locations.
- The data **dictionary** contains information about database objects such as tables, indexes, columns, data types, and views.
- **Data dictionary functions**
 - Names of all the database tables and their schemas.
 - Details about all the tables in the database, such as their owners, their security constraints, when they were created etc.
 - Physical information about the tables such as where they are stored and how.
 - Table constraints such as primary key attributes, foreign key information etc.
 - Information about the database views that are visible.



Data Dictionary Example

Data item	Metadata					
Name	Type	Length	Min	Max	Description	Source
Course	Alphanumeric	30			Course id	Academic
Section	Integer		1	5	Section no	Registrar
Semester	Integer	2	1	8	Semester no	Registrar
Name	Alphabetic	30			Student name	Student
CNIC	Integer	15			Computerized National Identity Card	NADRA
GPA	Decimal	2	0.0	4.0	Grade Point Average	ERP



Data Models

WHAT IS MODEL?

*"A representation of **real world objects, events** and their **association** is called a model"*

Benefit: It helps in understanding the complexities of real-world environments.

WHAT IS DATAMODEL?

*"A collection of concepts to **describe** and **manipulate** data, **relationship** b/w data and **constraints** on the data"*

Benefit: It helps in understanding the complexities of data and finding relevant information from it.

Data Models

PARTS OF DATAMODEL

1

STRUCTURAL PART

*“It consists of set of rules, that specify **how database can be developed.**”*

2

MANIPULATIVE PART

*“It defines the types of **operations that can be performed on data**, It includes the operations like updating/retrieving data or changing database structure.”*

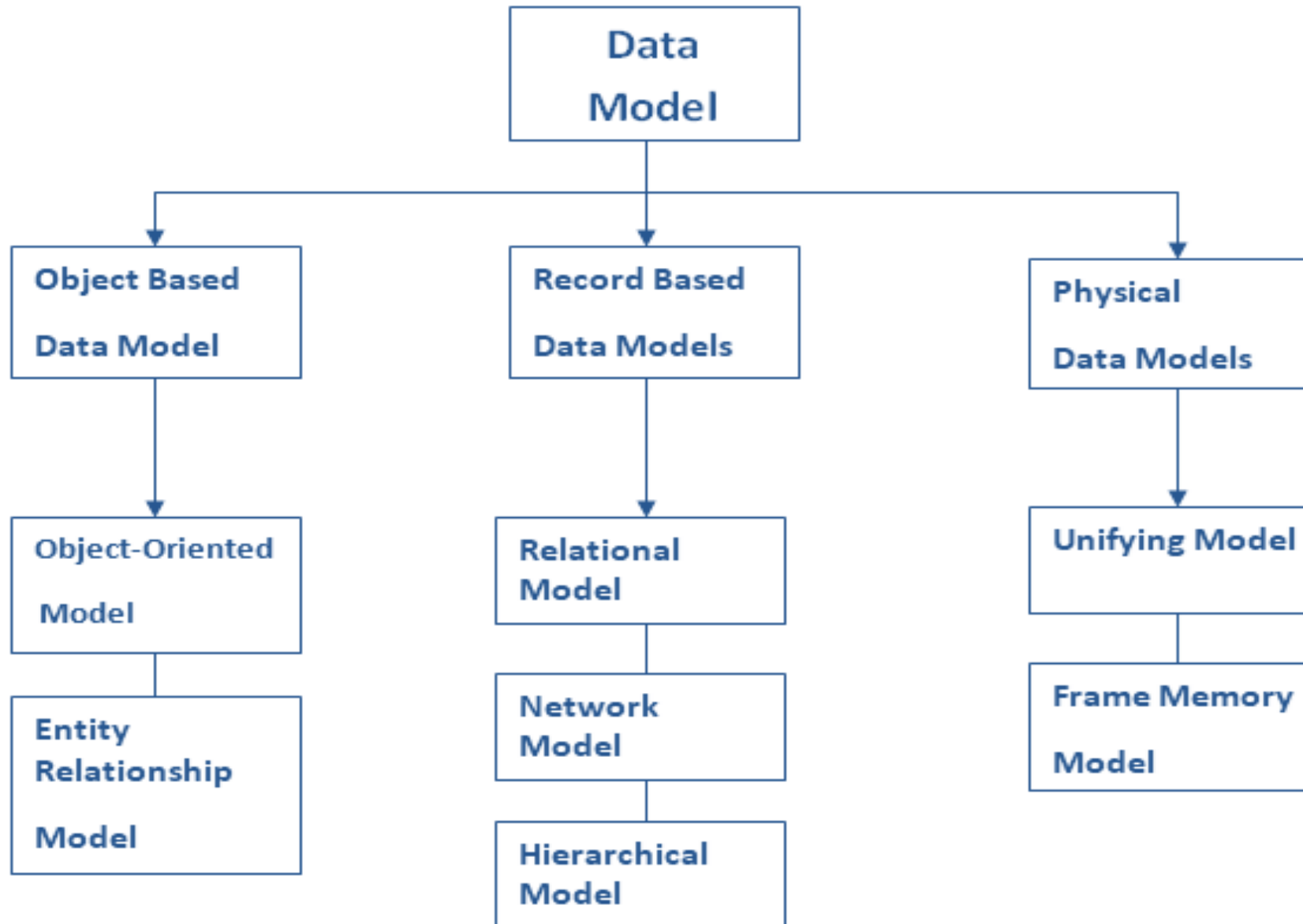
3

INTEGRITY RULES PART

*“It includes the rules that **ensure the accuracy** in the database.”*



Types of Data Models



Types of Data Models

OBJECT DATA MODEL

*“Object based data models **use concepts like entities, attributes and relationships**”*

- | | |
|--------------------------|--|
| 1. <u>Entity</u> : | Person/Place/Thing/Event for which data is collected. |
| 2. <u>Relationship</u> : | An association between two or more entities. |
| 3. <u>Attributes</u> : | Characteristics or properties of one or more entities. |

RECORD DATA MODEL

*“Record based data models **are consist of fixed-format records** of difference record types. Each record type **define fix number of fields**, each field is of **fix length**.”*

PHYSICAL DATA MODEL

*“This model represents the **store procedures** such as **record structures, record ordering & access paths**.”*



Types Data Models

OBJECT BASED DATA MODELS

1

ENTITY RELATIONSHIP MODEL

*“This model is a **real world graphical approach to database design**. It is a high-level model that defines **data elements and their relationship** for a specified software system.”*

2

SEMANTIC OBJECT MODEL

*“Semantic object model is **logical representation of data** in an organization. It views the **entire system as collection of objects** related to one & another. These objects are **grouped into the classes and instances**.”*



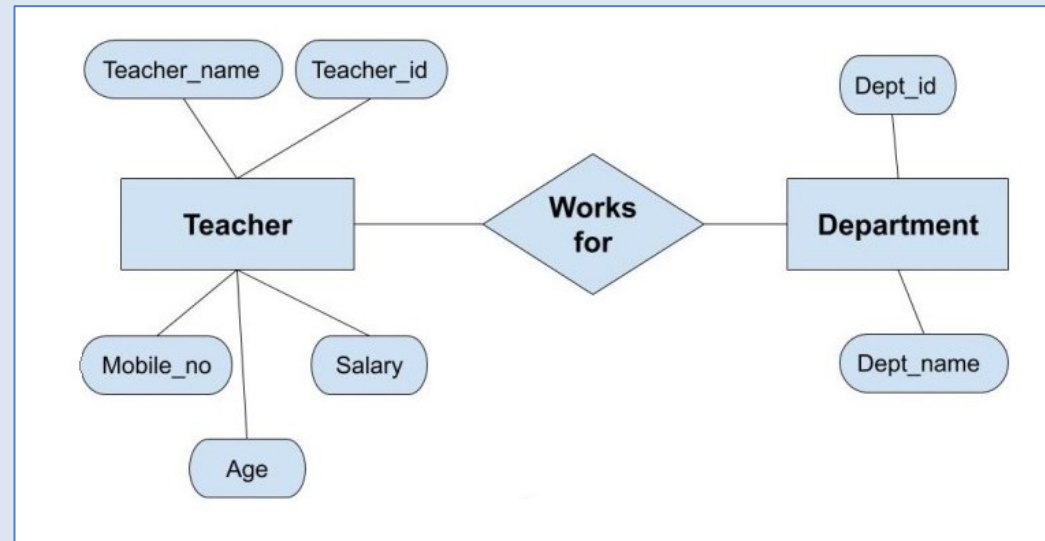
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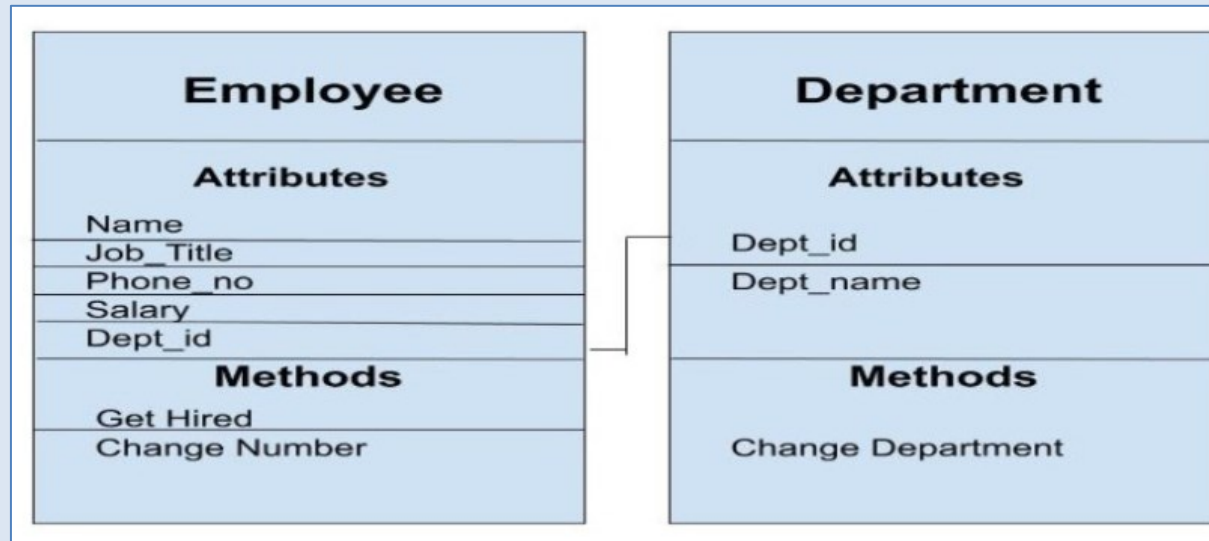
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Types Data Models

RECORD BASED DATA MODELS

1

HIRERCHICAL MODEL

*“In hierarchical data model, data is connected by **parent child relationship** which is **organized into a tree-like structure**. Related data records are **connected through nodes and pointers**.”*

2

NETWORK MODEL

*“This model was created to **resolve short comings of hierarchical** model by allowing **multiple parent child relationships**, in this model a child can be linked with **more than one parents**.”*

3

RELATIONAL MODELS

*“In relational models **data is stored in relations/tables**. Each table in database has **unique name** that identify its concepts, **table/relation** is also called **intersection of rows and columns**.”*



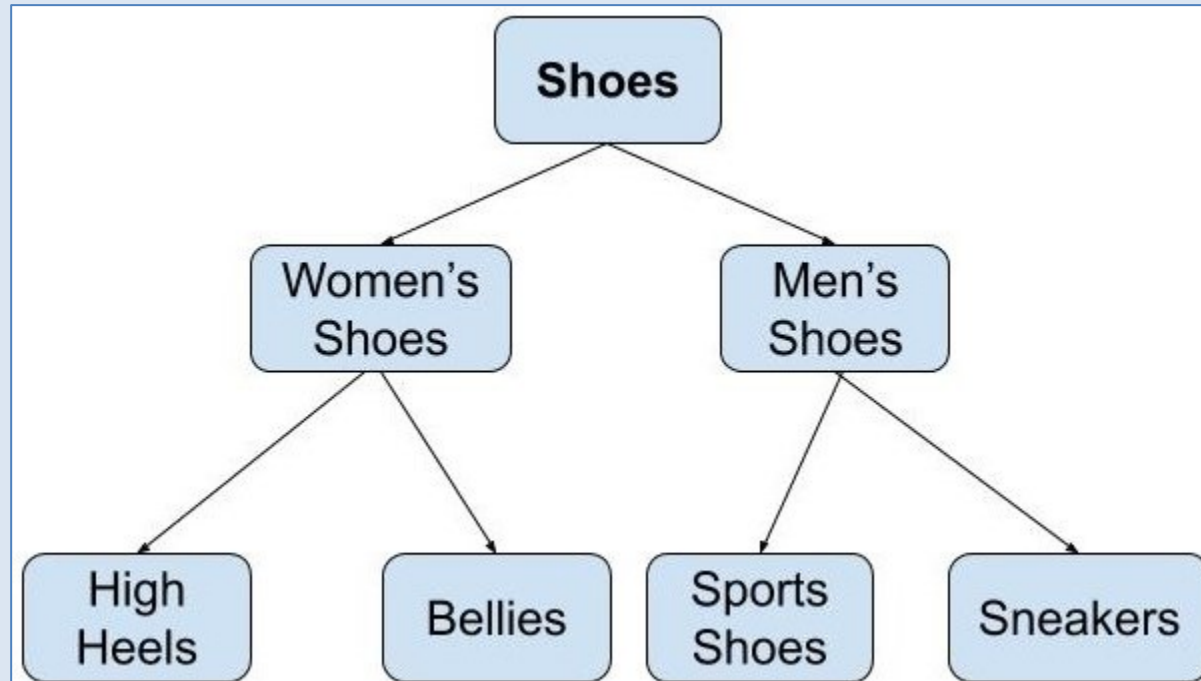
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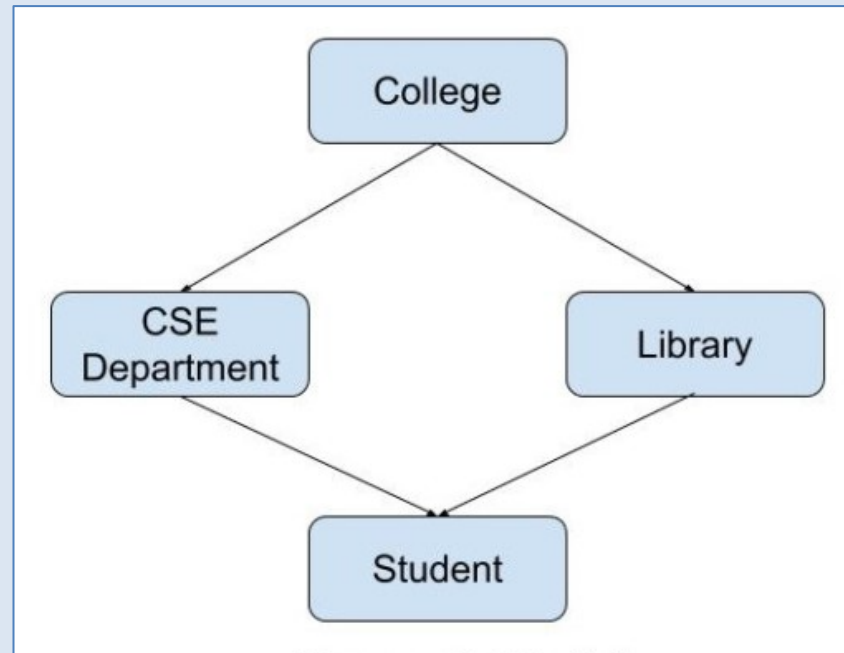
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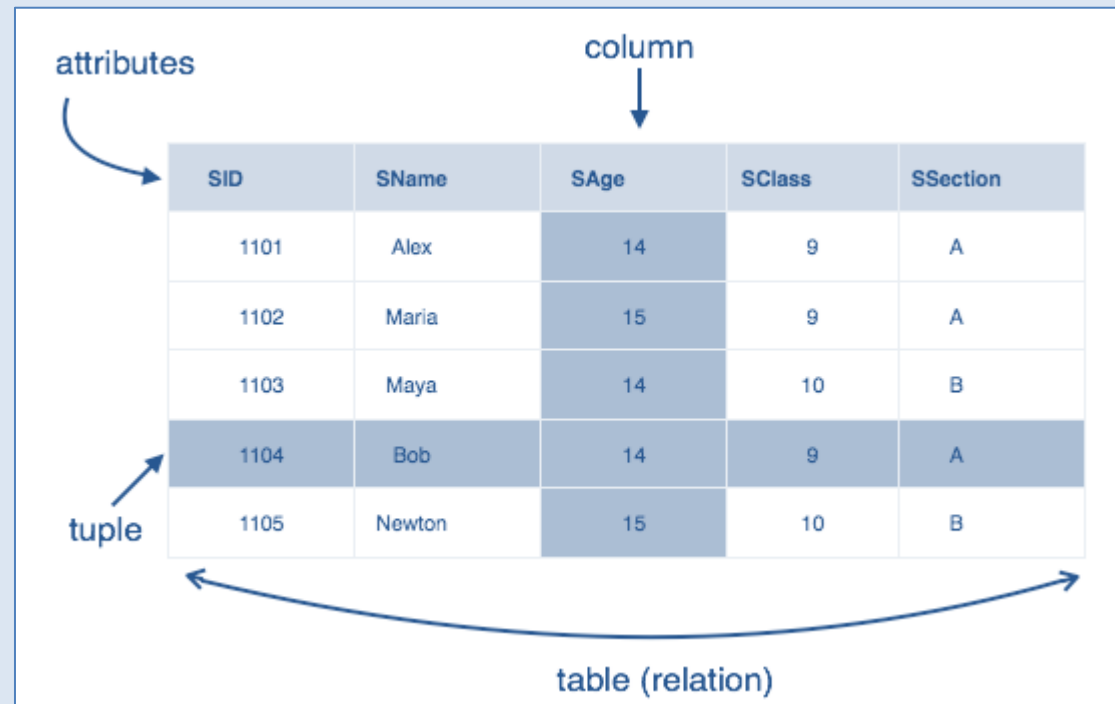
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Types Data Models

PHYSICAL DATA MODELS

1

UNIFYING MODEL

*“Unified data model **acts as a bridge** between two different data models / ecosystems. It builds foundation for **collection & correlation of data from multiple sources.**”*

2

FRAME MEMORY MODEL

*“Frame memory is a **virtual view of secondary storage** that can be implemented with reasonable overhead to support database record storage and accessing requirements.”*



Types Data Models

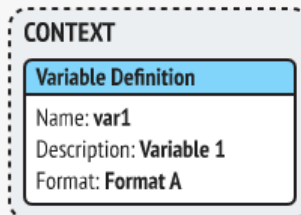
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Variable Reading/Writing



Reading

DATA TABLE

Represents current variable value
(matches **Format A**)

Writing

DATA TABLE

Represents new variable value
(matches **Format A**)

Types Data Models

PHYSICAL DATA MODELS

2

FRAME MEMORY MODEL

*“Frame memory is a **virtual view of secondary storage** that can be implemented with reasonable overhead to support database record storage and accessing requirements.”*

